

Lab # 02

Basic Syntax, Input/Output, Data Types **& Arithmetic Operations in C++**

Objective:

To understand the basic structure of a C++ program, perform input/output operations, use different data types, and apply arithmetic operators.

Theory:

1. Basic Syntax of C++

A c++ program follows a specific structure:

```
#include <iostream> ? used for input/output
using namespace std; ? allows use of standard library names
int main() ? main function (starting point of program)
{ } ? block of code
return 0; ? ends program
```

2. Input and Output in C++

```
cout ? used to display output
cin ? used to take input from user
```

Example:

```
cout << "Hello World";
cin >> variable;
```

3. Data Types in C++

Data types define the type of data a variable can store:

Data Type

Description

Example:

- **Int**
Integer values
- **int a = 5;**
- **Float**

- Decimal numbers
`float b = 3.5;`
- **Double**
Large decimal numbers
`double x = 10.25;`
- **Char**
Single character
`char c = 'A';`
- **String**
Collection of characters
`string name = "Ali";`

4. Arithmetic Operators

Used to perform mathematical operations:

Operator

Meaning

Example

- **+**

Addition
`a + b`
- **-**
Subtraction
`a - b`
- *****
Multiplication
`a * b`
- **Division**
`a / b`
- **%**
Modulus (remainder)
`a % b`

Program:

```

#include <iostream> // input aur output ke liye library
using namespace std; // std ko baar baar likhne se bachne ke liye

int main() {
    int a, b; // do integer variables banaye

    cout << "Enter two numbers: "; // user se numbers mang rahe hain
    cin >> a >> b; // user ke diye gaye numbers ko store kar rahe hain

    cout << "Addition: " << a + b << endl; // dono numbers ka jama (plus)
    cout << "Subtraction: " << a - b << endl; // pehle se doosra minus
    cout << "Multiplication: " << a * b << endl; // dono ka multiply
    cout << "Division: " << a / b << endl; // pehla number doosre se divide
    cout << "Modulus: " << a % b << endl; // remainder (baqi) nikalna

    return 0; // program khatam
}

```

Explanation:

- Two integers a and b are declared.
- User inputs values using cin.
- Arithmetic operations are performed.
- Results are displayed using cout.

Sample Output:

Enter two numbers: 10 5

Addition: 15

Subtraction: 5

Multiplication: 50

Division: 2

Modulus: 0

Conclusion:

In this lab, we learned:

- Structure of a C++ program
- How to take input and display output
- Different data types
- Use of arithmetic operators

Precautions:

- Always use semicolon ; at end of statements
- Avoid division by zero
- Use correct data types.

Lab Tasks:

(1):

```
#include<iostream> // input aur output ke liye library
using namespace std; // std ko use karne ke liye

int main() {
    int a = 5; // integer variable 'a' banaya aur usmein 5 store kiya
    float b = 2.16; // float variable 'b' banaya aur usmein 2.16 rakha
    double c = 8.5423; // double variable 'c' banaya aur usmein 8.5423 rakha

    // 'a' ki value, size aur memory address print kar rahe hain
    cout << "value of a " << a << endl; // value show karna
    cout << "size of a " << sizeof(a) << endl; // size bytes mein batana
    cout << "location of a " << &a << endl; // memory address dikhana

    // 'b' ki value, size aur memory address print kar rahe hain
    cout << "value of b " << b << endl; // value show karna
    cout << "size of b " << sizeof(b) << endl; // size batana
    cout << "location of b " << &b << endl; // memory address dikhana

    // 'c' ki value, size aur memory address print kar rahe hain
    cout << "value of c " << c << endl; // value show karna
    cout << "size of c " << sizeof(c) << endl; // size batana
    cout << "location of c " << &c << endl; // memory address dikhana

    return 0; // program khatam
}
```

(2):

```

#include<iostream> // input aur output ke liye library
using namespace std; // standard namespace use karne ke liye

int main()
{
    float tempcel, tempfah; // do variables banaye: ek Celsius ke liye aur ek Fahrenheit ke liye

    cout << "Enter temperature in Celsius :" << endl; // user se Celsius temperature input lene ke liye
    cin >> tempcel; // user ka diya hua temperature store karna

    // Celsius ko Fahrenheit mein convert karne ka formula:  $F = (C * 9/5) + 32$ 
    tempfah = (tempcel * 9 / 5) + 32;

    cout << "Temperature in Fahrenheit = " << tempfah << endl; // Fahrenheit value display karna

    return 0; // program yahan khatam hota hai
}

```

(3):

```

#include <iostream>
using namespace std;

int main()
{
    float pi = 3.14;
    // Yeh ek float type ka variable 'pi' banaya gaya hai jismein 3.14 store kiya gaya hai

    cout << "Value of pi is: " << 3.14 << endl;
    // Yeh line screen par "Value of pi is: 3.14" print karti hai

    return 0;
    // Program yahan khatam hota hai aur 0 return karta hai (successful execution)
}

```
